

KILLARA HIGH SCHOOL



YEAR 12 MATHEMATICS EXTENSION 2

2008

HSC Assessment 1

Time Allowed: 70 minutes

Instructions:

- Attempt all questions
- Start each question on a new page
- Show ALL necessary working
- Marks may be deducted for untidy or badly arranged work
- Board approved calculators may be used.

Outcomes Assessed:

- E1 Appreciates the creativity, power and usefulness of mathematics to solve a broad range of problems
- E3 Uses the relationship between algebraic and geometric representations of complex numbers and of conic sections.
- E9 Communicates abstract ideas and relationships using appropriate notation and logical argument.

This assessment task constitutes 10% of the final HSC course Assessment.

Question 1 (15 Marks) START A NEW PAGE

(a). If $z = 3 + \sqrt{2}i$, find in the form $a + ib$, **5**

(i) $\bar{z}$ (ii) $z + \bar{z}$ (iii) $z\bar{z}$ (iv) z^{-1}

(b) (i) Find the $\sqrt{16 - 30i}$. **2**

(ii) Hence, or otherwise solve the equation $z^2 - (1 - i)z + 7i - 4 = 0$
in the form $a + ib$ **3**

(c) By writing each factor in the modulus-argument form, simplify **5**

$$(\sqrt{3} + i)^6 \div (1 - i)^4$$

Question 2 (15 Marks) START A NEW PAGE

(a) Sketch and describe the loci defined by the following:

(i) $|z - 4| = |z + 2i|$ **2**

(ii) $z\bar{z} = z + \bar{z}$ **4**

(b) $1 - 2i$ is one of the roots of $x^2 + (1 + i)x + k = 0$

(i) State the other root. **2**

(ii) Find the value of k . **2**

(c) Given that 1 , ω and ω^2 are the complex roots of unity.
Evaluate:

(i) $1 + \omega + \omega^2$ **2**

(ii) $(1 + \omega^2)^{12}$ **3**

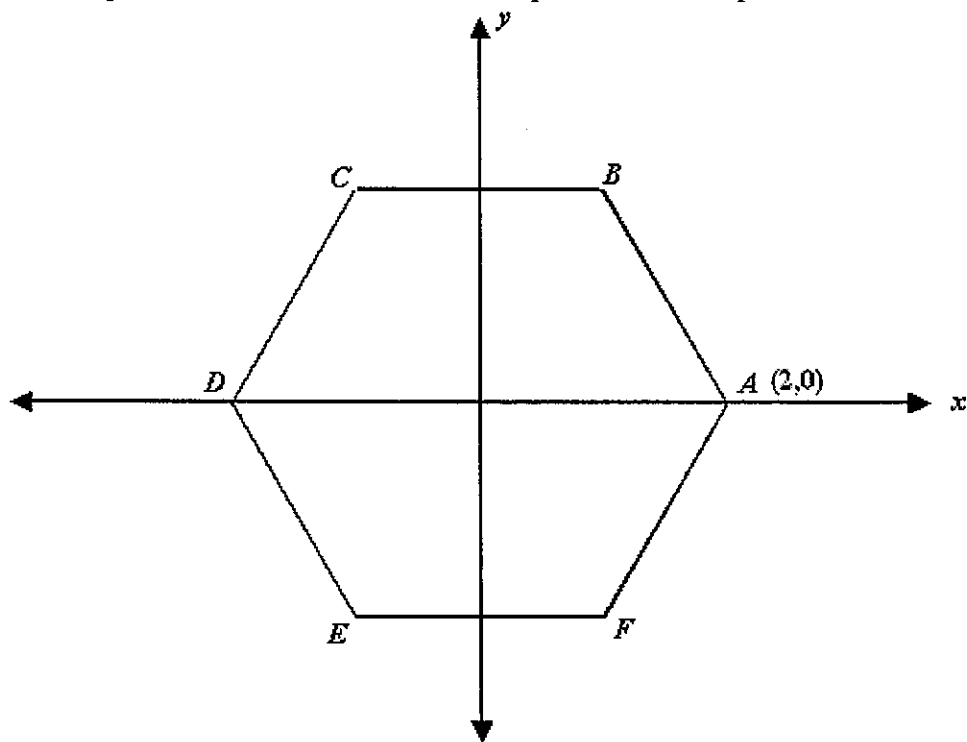
Question 3 (20 Marks) START A NEW PAGE

(a) Find the locus of z , if $W = \frac{z-i}{z-2}$ is purely imaginary. 4

(b) (i) Find the five roots of $Z^5 = 1$. 3

(ii) Hence, express $Z^4 + Z^3 + Z^2 + Z + 1$ as product of two quadratic factors. 3

(c)



$ABCDEF$ is a regular hexagon drawn on the Argand diagram with the vertex A at the point $(2, 0)$. O is the centre of the hexagon.

(i) Copy the diagram into your answer book.

(ii) On your diagram show the region within the hexagon in which both the inequalities $|z| \geq 1$ and $-\frac{\pi}{3} \leq \arg z \leq \frac{\pi}{3}$ are satisfied. 2

(iii) Find, in the form $|z - c| = R$, the equation of the circle through the points O, B and F . and sketch it on the Argand diagram. 2

(iv) Find the complex numbers represented by the points C and E 3

(v) The hexagon is rotated anticlockwise about the origin through an angle of $\frac{\pi}{4}$. 3
Express in the form $r(\cos \theta + i \sin \theta)$ the complex numbers represented by the new positions of C and E .

End of paper

Question 1

KHS. EXT 2

Term 4.

Assessment 1

2008.

① $z = 3 + \sqrt{2}i$

(i) $\bar{z} = 3 - \sqrt{2}i$ ✓ ①

(ii) $z + \bar{z}$

$= 3 + \sqrt{2}i + 3 - \sqrt{2}i$

$= 6$ ✓ ①

(iii) $z\bar{z}$

$= (3 + \sqrt{2}i)(3 - \sqrt{2}i)$

$= 9 + 2 = 11$ ✓ ①

(iv) $z^{-1} = \frac{1}{z}$

$= \frac{1}{3 + \sqrt{2}i} = \frac{3 - \sqrt{2}i}{(3 + \sqrt{2}i)(3 - \sqrt{2}i)}$ ✓

$= \frac{3 - \sqrt{2}i}{11}$ ✓ ②

(b) (i) $\sqrt{16 - 30i}$

$p = 5 \times 3$

$s = 16$

$f = 5 \quad 3i$

$16 - 30i = (5 - 3i)^2$

$\therefore \sqrt{16 - 30i} = \sqrt{(5 - 3i)^2}$ ②

$= \pm(5 - 3i)$ ✓

(ii) $z^2 - (1 - i)z + 7i - 4 = 0$

$z = \frac{(1 - i) \pm \sqrt{(1 - i)^2 - 4 \times 1(7i - 4)}}{2}$

$= \frac{(1 - i) \pm \sqrt{1 - 2i - 1 - 28i + 16}}{2}$ ✓

$= \frac{(1 - i) \pm \sqrt{16 - 30i}}{2}$

$= \frac{(1 - i) \pm (5 - 3i)}{2}$ ✓

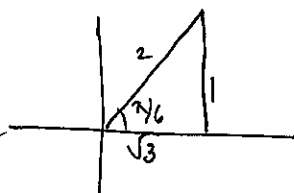
$= \frac{1 - i + 5 - 3i}{2}, \frac{1 - i - 5 + 3i}{2}$

$= \frac{6 - 4i}{2}, \frac{2i - 4}{2}$

$= 3 - 2i, -2 + i$ ✓ ③

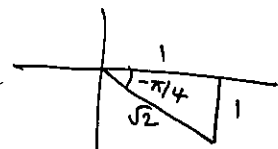
(c) $\sqrt{3} + i =$

$= 2 \cos \frac{\pi}{6}$ ✓



$1 - i =$

$\sqrt{2} \cos(-\frac{\pi}{4})$ ✓



$\frac{(\sqrt{3} + i)^6}{(1 - i)^4} = \frac{(2 \cos \frac{\pi}{6})^6}{(\sqrt{2} \cos(-\frac{\pi}{4}))^4}$ ✓

$= \frac{2^6 \cos(\frac{\pi}{6} \times 6)}{2^2 \cos(-\frac{\pi}{4} \times 4)}$ ✓

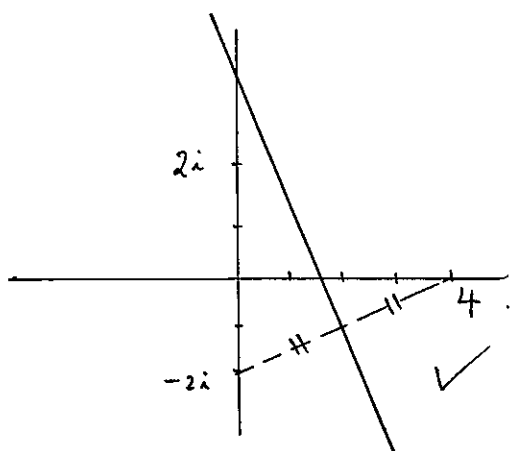
$= \frac{2^6 \cos \pi}{2^2 \cos(-\pi)} = \frac{2^6 \times -1}{2^2(-1)}$ ⑤

$= 2^4 = 16$ ✓

Question 2

(2)

(a) (i) $|z-4| = |z+2i|$



The locus is the perpendicular bisector of the line joining $(4, 0)$ and $(0, -2i)$. ✓ (2)

$$|x+iy-4| = |x+iy+2i|$$

$$|x-4+iy| = |x+i(y+2)|$$

$$(x-4)^2 + y^2 = x^2 + (y+2)^2$$

$$x^2 - 8x + 16 + y^2 = x^2 + y^2 + 4y + 4$$

$$8x + 4y - 12 = 0$$

$$\cancel{4x} \quad 2x + y - 3 = 0$$

(ii) $z\bar{z} = z + \bar{z}$ let $z = x+iy$.

$$(x+iy)(x-iy) = x+iy + x-iy$$

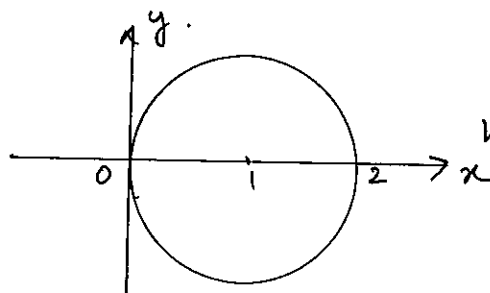
$$x^2 + y^2 = 2x$$

$$x^2 - 2x + y^2 = 0$$

$$x^2 - 2x + 1 + y^2 = 1$$

$$(x-1)^2 + y^2 = 1 \quad \checkmark$$

The locus is a circle with centre $(1, 0)$ and radius 1. ✓



(4)

(b) $x^2 + (1+i)x + k = 0$.

(i)

$1-2i$ is a root.

Let the other root be α .

$$\therefore \text{Sum of roots} = 1-2i + \alpha = -\frac{(1+i)}{1} \quad \checkmark$$

$$\alpha = -1-i-1+2i = -2+i \quad \checkmark \quad (2)$$

(ii) Product of roots = $\frac{c}{a}$.

$$(1-2i)(-2+i) = \frac{k}{1} \quad \checkmark$$

$$\therefore k = -2+i+4i+2 = 5i \quad \checkmark \quad (2)$$

(c) Consider the roots of the equation $z^3 = 1$.

$1, \omega, \omega^2$ are the three roots of $z^3 = 1$.

$$\therefore \text{Sum of roots} = -\frac{b}{a} = -\frac{\text{Coefficient of } z^2}{\text{Coefficient of } z^3} = \frac{0}{1} = 0$$

$$\therefore 1 + \omega + \omega^2 = 0 \quad \checkmark$$

③

$$(ii) (1 + \omega^2)^{12}$$

$$1 + \omega + \omega^2 = 0, \quad \omega^3 = 1$$

$$\therefore 1 + \omega^2 = -\omega \quad \checkmark$$

$$\therefore (1 + \omega^2)^{12} = (-\omega)^{12}$$

$$= (\omega^3)^4 \quad \checkmark$$

$$= 1^4$$

$$= \underline{\underline{1}} \quad \checkmark$$

③

Question 3

$$(a) \quad \omega = \frac{z-i}{z-2}$$

$$\text{let } z = x + iy$$

$$\therefore \omega = \frac{x + iy - i}{x + iy - 2}$$

$$= \frac{x + (y-1)i}{(x-2) + iy} \quad \checkmark$$

$$= \frac{x + (y-1)i}{(x-2) + iy} \times \frac{(x-2) - iy}{(x-2) - iy} \quad \checkmark$$

$$= \frac{x(x-2) - xyi + (y-1)(x-2)i + y(y-1)}{(x-2)^2 + y^2}$$

$$= \frac{x^2 - 2x + y^2 - y + i(-xy + (y-1)(x-2))}{(x-2)^2 + y^2} \quad \text{is}$$

purely imaginary.

$$\therefore \operatorname{Re}(\omega) = 0.$$

$$(x-2)^2 + y^2 \neq 0.$$

$$\therefore x^2 - 2x + y^2 - y = 0 \quad \checkmark$$

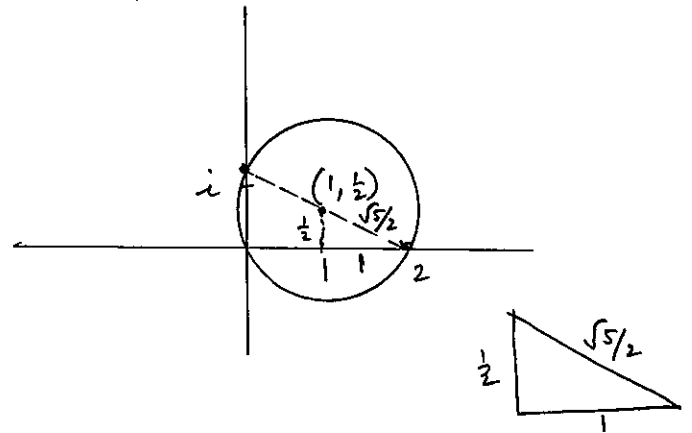
$$(x-1)^2 + (y-\frac{1}{2})^2 = 1 + \frac{1}{4}$$

$$(x-1)^2 + (y-\frac{1}{2})^2 = \frac{5}{4}. \quad \textcircled{4}$$

$\therefore$ locus is a circle with $\checkmark$
Centre $(1, \frac{1}{2})$ and radius $\frac{\sqrt{5}}{2}$.

OR

$$\omega = \frac{z-i}{z-2} \text{ is purely imaginary.}$$



Mid-point of $(0, 1)$ and $(2, 0)$
 $= (1, \frac{1}{2})$ and $r = \frac{\sqrt{5}}{2}.$

$\therefore$ locus is a circle with
centre $(1, \frac{1}{2})$ and radius $\frac{\sqrt{5}}{2}.$

$$(x-1)^2 + (y-\frac{1}{2})^2 = \underline{\underline{\frac{5}{4}}}.$$

(4)

(b) (i) $z^5 = 1$

$r = 1.$

let $z = \text{cis } \theta$.

$z^5 = \text{cis } 5\theta = 1 + 0i$

Comparing real and imaginary parts,

$\cos 5\theta = 1 \quad \sin 5\theta = 0. \checkmark$

$\therefore 5\theta = 2k\pi + 0$

$\therefore \theta = \frac{2k\pi}{5}, k = 0, \pm 1, \pm 2. \checkmark$

$k=0, z_1 = \text{cis } 0 = 1$

$k=1, z_2 = \text{cis } \frac{2\pi}{5}$

$k=-1, z_3 = \text{cis } \left(-\frac{2\pi}{5}\right) = \overline{z_2}$

$k=2, z_4 = \text{cis } \frac{4\pi}{5}$

$k=-2, z_5 = \text{cis } \left(-\frac{4\pi}{5}\right) = \overline{z_4}$

(b) (ii)

$z^5 - 1 = (z-1)(z^4 + z^3 + z^2 + z + 1)$

 $\therefore z_2, z_3, z_4, z_5$ are the roots $z^4 + z^3 + z^2 + z + 1 = 0.$ The quadratic with z_2, z_3 as roots.

$$\left\{ \begin{array}{l} \text{Sum of roots } z_2 + z_3 = z_2 + \overline{z_2} \\ \quad \quad \quad = 2 \cos \frac{2\pi}{5} \\ z_2 z_3 \text{ Product of roots} \\ \quad \quad \quad = z_2 \overline{z_2} = \cos^2 \frac{2\pi}{5} + \sin^2 \frac{2\pi}{5} = 1 \end{array} \right.$$

 $\therefore$ Quadratic is

$z^2 - 2 \cos \frac{2\pi}{5} z + 1 \quad \checkmark$

Similarly the other quadratic with z_4, z_5 as roots is

$z^2 - 2 \cos \frac{4\pi}{5} z + 1.$

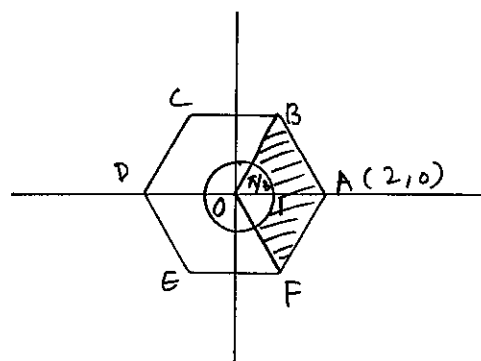
(3)

$\therefore z^4 + z^3 + z^2 + z + 1 =$

$\underbrace{(z^2 - 2 \cos \frac{2\pi}{5} z + 1)}_{\checkmark} (z^2 - 2 \cos \frac{4\pi}{5} z + 1).$

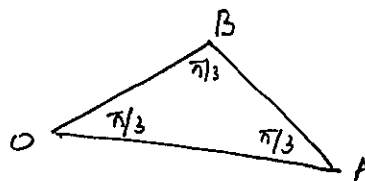
(c)

(i)

(ii) $|z| \geq 1$ outside the circleCorrect region shown $\checkmark$ (2)

$-\pi/3 \leq \arg(z) \leq \pi/3$ Correct region $\checkmark$

(iii)

~~OB = OA~~ $\triangle OBA$ is equilateral

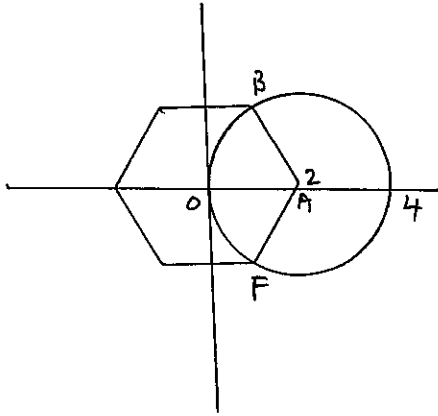
$\therefore OB = OA = OF = 2$

 $\therefore$ equation of the circle

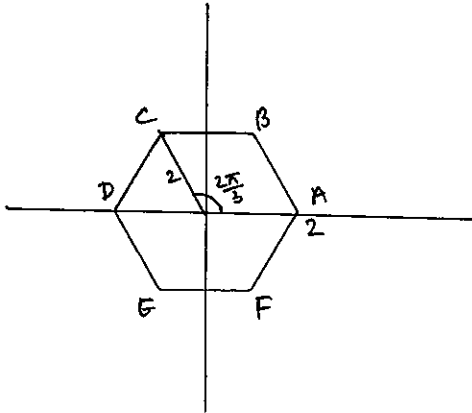
$|z - 2| = 2 \quad \checkmark \quad \boxed{1}$

See diagram below.

(5)



(iv)



Point A(2,0)

$$OA = OC$$

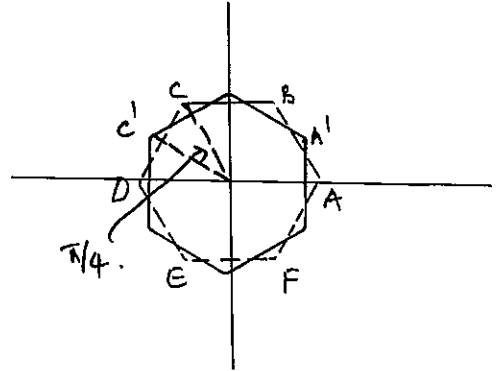
$\vec{OC}$ is obtained by rotating $\vec{OA}$ through $\frac{2\pi}{3}$ radians.

$$\therefore \vec{OC} = 2 \operatorname{cis}\left(\frac{2\pi}{3}\right) \checkmark$$

$$= 2\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = -1 + \sqrt{3}i \checkmark$$

$$\therefore E = -1 - \sqrt{3}i \checkmark \quad (3)$$

(v)



$$C = 2 \operatorname{cis}\left(\frac{2\pi}{3}\right)$$

$$\therefore C' = 2 \operatorname{cis}\left(\frac{2\pi}{3} + \frac{\pi}{4}\right)$$

$$= 2 \operatorname{cis}\left(\frac{11\pi}{12}\right)$$

$$= 2\left[\cos\left(\frac{11\pi}{12}\right) + i \sin\left(\frac{11\pi}{12}\right)\right] \checkmark$$

$$E = 2 \operatorname{cis}\left(-\frac{2\pi}{3}\right) \checkmark$$

$$\therefore E' = 2 \operatorname{cis}\left(-\frac{2\pi}{3} + \frac{\pi}{4}\right)$$

$$= 2 \operatorname{cis}\left(-\frac{5\pi}{12}\right)$$

$$* \vec{E'} = 2\left[\cos\left(\frac{5\pi}{12}\right) - i \sin\left(\frac{5\pi}{12}\right)\right] \checkmark$$